Twitter and the Politics of the Federal Deficit

Results (Draft)

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! Important

Writing sample disclaimer. This PDF contains a draft of the *Results* section from the working paper
Twitter and the Politics of the Federal Deficit. The abstract is included below for context.

0.1 Abstract (for context)

A long-held belief in American politics is that elected officials' concern about the federal deficit is mainly a function of who holds power. Amid deepening polarization, elevated public debt, and the shift of political communication online, it is vital—for voters, policymakers, and public-knowledge institutions—to understand empirically how deficit rhetoric operates in these spaces. I analyze ~3.6 million tweets from official congressional accounts (June 2017–January 2023), identify deficit-related posts using a two-tier keyword approach, and estimate member–month logistic models linked to leadership roles and legislative windows. My research confirms that deficit rhetoric is rare but fluctuates greatly depending on two contexts: relative institutional power (ie, minority / majority party), and legislative periods (ie, Tax Cuts and Jobs Act, Inflation Reduction Act). These findings contribute to understanding the modern incentive structures and patterns of U.S. political communication strategies, illustrating that deficit rhetoric is less a barometer of underlying fiscal conditions than a tool used to serve political and partisan interests.

1 Macro Trends

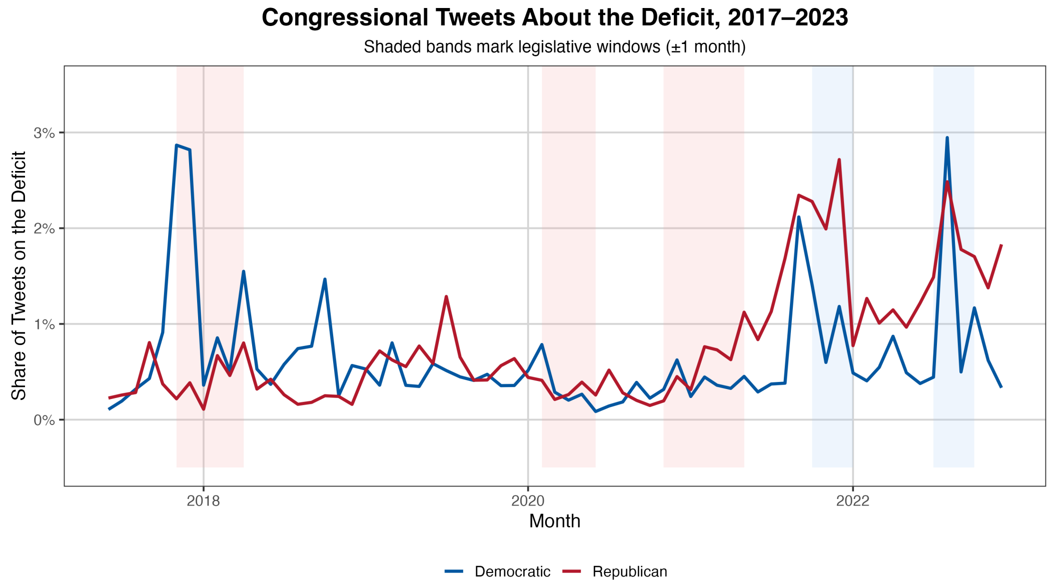

Figure 1: Share of congressional tweets about the deficit

Figure 1 shows the monthly share of congressional deficit tweets, by party, from June 2017 to January 2023. Consistent with best practices in political science, this paper will represent the Republican Party with red shading and the Democratic Party with blue. Shaded areas indicate legislative periods – defined as the month before, during, and after the passage of a fiscally significant bill. Legislative periods exclude small appropriations, procedural votes, and commemorations.

The baseline deficit tweeting share for both parties is low, generally staying under 1% across the six years, with notable accelerations occurring around legislative periods. Democratic tweeting patterns remain relatively consistent across the six years, with three notable peaks, all during legislative periods: December 2017 (Tax Cuts and Jobs Act), October 2021 (Infrastructure Investment and Jobs Act), and August 2022 (Inflation Reduction Act, CHIPS & Science Act). Republicans, however, demonstrate consistently higher deficit tweeting rates as the minority party, as illustrated in the peaks (2022, 2023) compared to relatively low rates from 2017-2021.

2 Economic Indicators

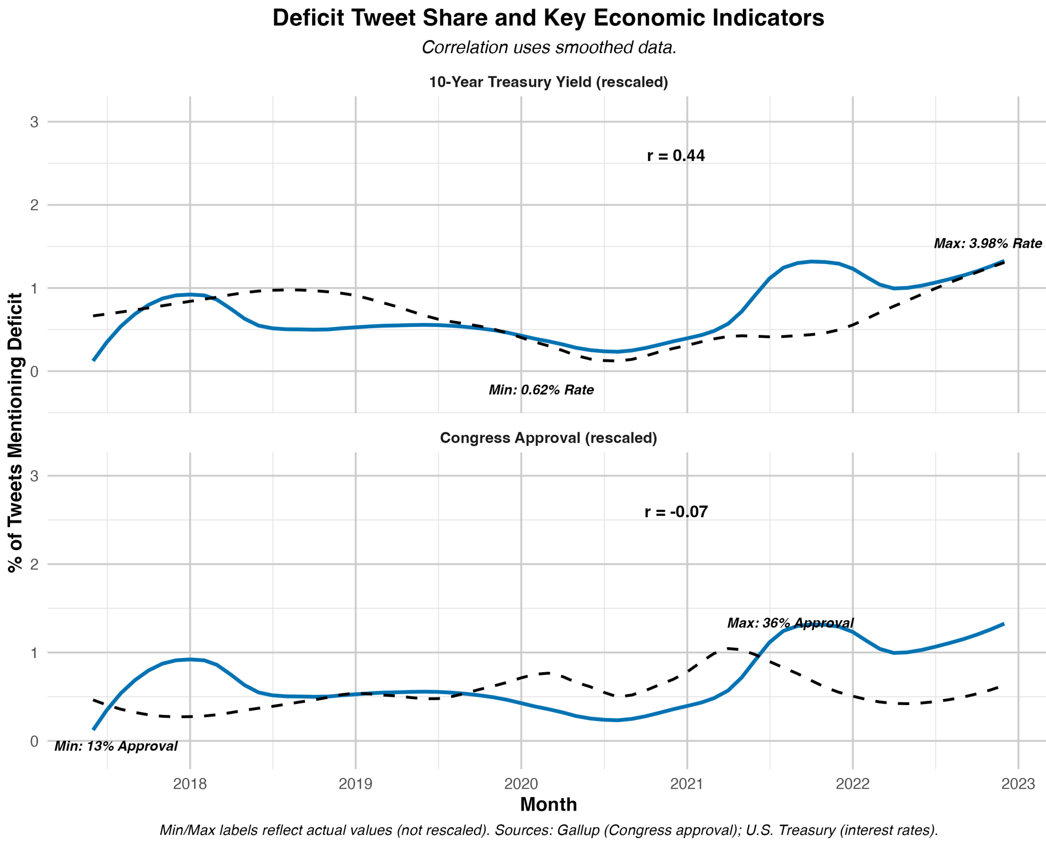

Figure 2: Deficit tweet share vs 10-year Treasury yield and congressional approval; rescaled.

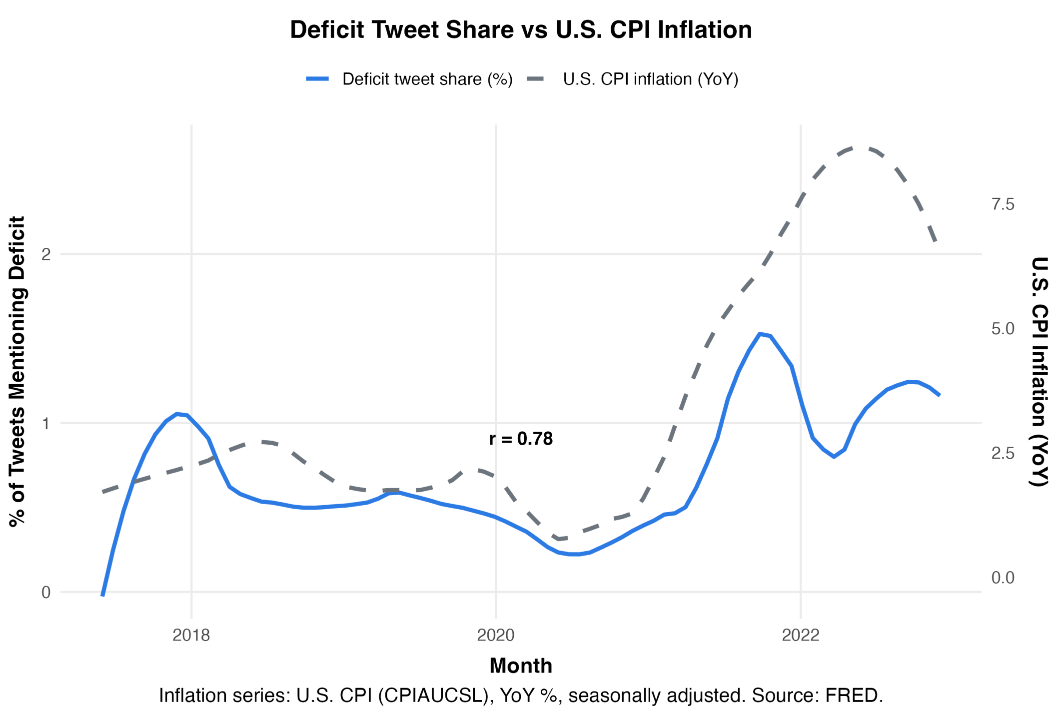

Figure 3: Deficit tweet share vs. U.S. CPI inflation.

Figure 2 plots deficit tweet share against the 10-year Treasury yield and congressional approval, both rescaled for comparability. The small correlation coefficients (0.44, -.07) suggest no discernible linear relationship. In contrast, Figure 3 – which plots deficit tweeting against US CPI inflation, reveals a relatively strong positive relationship, as indicated by its large r-value (.78). We will explore several plausible explanations for this correlation in the discussion section.

3 Member-Level Patterns

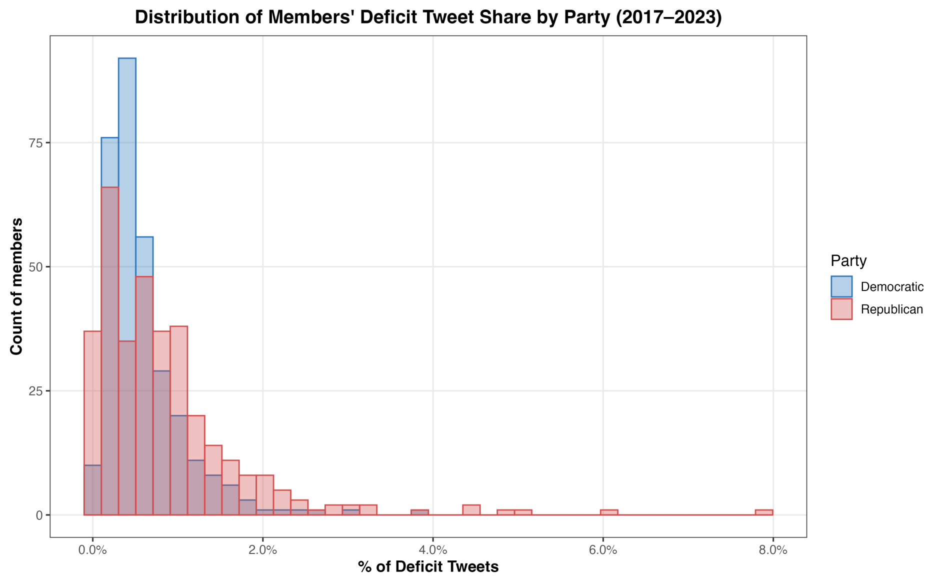

Figure 4: Share of deficit tweets by party (member-level distribution).

Deficit tweeting is uncommon across both Deficit tweeting is uncommon across both parties, with the vast majority of members posting fewer than **(1.5%)** of their tweets on the deficit. Democrats display a compact distribution: most fall below **(0.6%)** with no Democratic member exceeding **(2.5%)**. Republicans show greater variability, as illustrated by their distribution's long right tail, which reaches **(4–8%).**

Quartiles, Median, and Range of Members' Deficit Tweet Share

By party and power context (members with ≥ **500** total tweets)

	Party	N	Min	Q1	Median	Q3	Max	Range
In Power	Democratic	317	0.0%	0.2%	0.5%	0.8%	4.9%	0.0% – 4.9%
	Republican	344	0.0%	0.1%	0.2%	0.5%	6.7%	0.0% – 6.7%
Out of Power	Democratic	317	0.0%	0.2%	0.4%	0.6%	3.9%	0.0% – 3.9%
	Republican	344	0.0%	0.0%	1.0%	1.8%	8.9%	0.0% – 8.9%

Values shown as percent of member's tweets that mention the deficit. Range shown as *min – max*.

Figure 7: Distribution of deficit tweets by power (table).

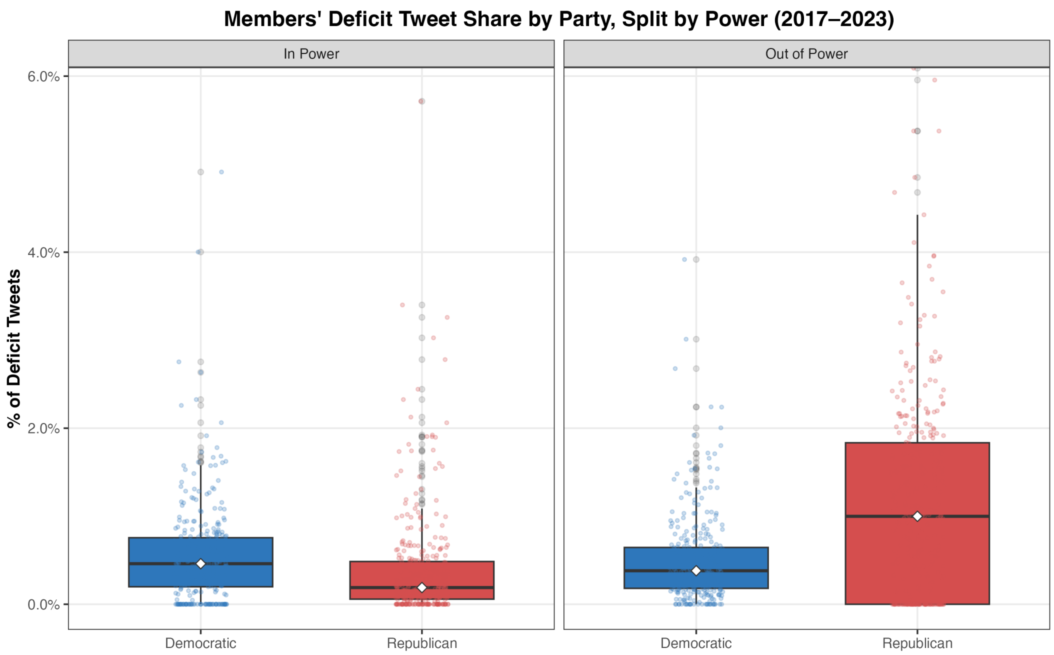

Figure 5: Distribution of deficit tweets by power (boxplot).

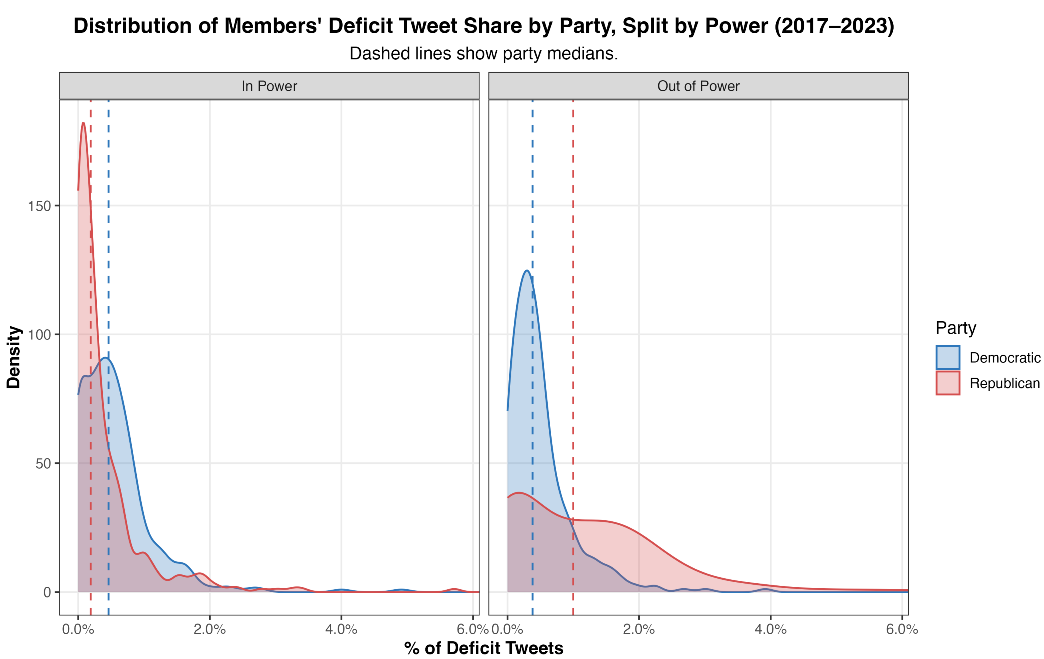

Figure 6: Distribution of deficit tweets by power (density plot).

As illustrated by the three figures above, Republicans' significant right skew stems largely from their tweeting while out of power. For instance, when moving between these contexts, the GOP median tweet share increases five-fold **(0.2% to 1.0%)** and the inter-quartile range from **(0.6%)** to **(1.8%)**. Democrats, by contrast, remain relatively stable across contexts, with only a modest increase in spread and averages when in power.

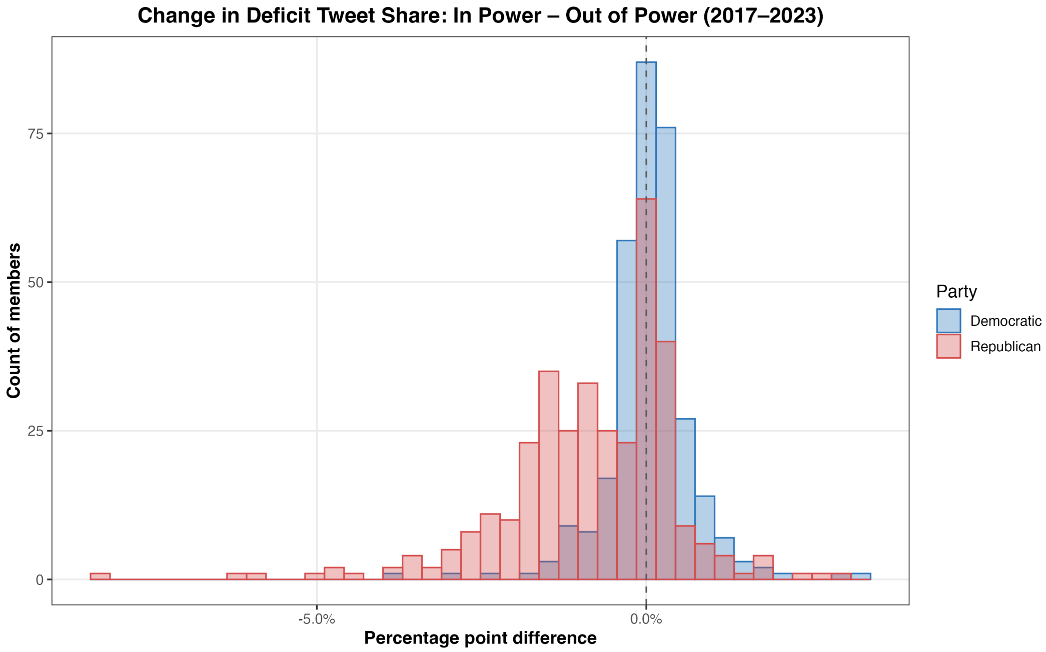

Figure 8: In–out shifts in deficit tweeting (histogram).

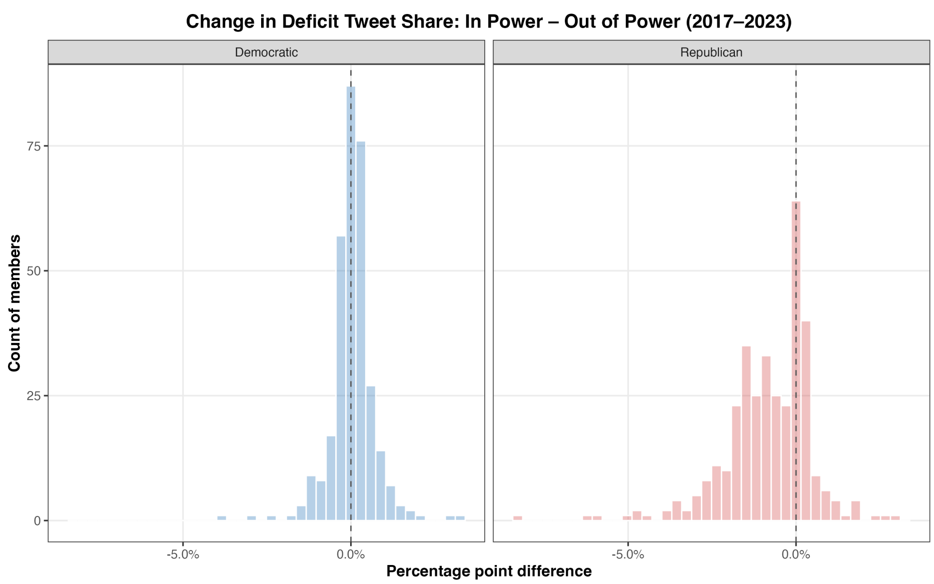

Figure 9: Faceted in–out shifts in deficit tweeting (histograms by subgroup).

Histograms of the change in deficit tweeting further highlight these asymmetries. Republicans' distribution is skewed negative, meaning the party tweets at higher frequencies when out of power. Both figures highlight this trend as the bulk of the GOP falls between **(–5.0%)** and **(+1%)**. Democrats, however, are roughly distributed around zero, indicating consistent levels of deficit tweeting across power status contexts.

These distributions demonstrate three points. First, deficit tweeting is relatively rare for both parties, with average levels rarely exceeding **(+1.5%)**. Second, most outliers – the members who commit **(4-8%)** of their tweets to deficits – are exclusively Republicans. Third, the GOP systematically increases its tweet share when out of power, whereas Democrats show little comparable shift.

4 Leadership Analysis

Deficit Tweeting Rates by Congressional Leaders (2017–2023)

Includes only leaders who served during this period

Leadership Role	Party	Total Tweets	Debt Tweets	% Deficit-Related
Chuck Schumer				
Senate Majority Leader	Democratic	4173	167	4.00
Senate Minority Leader	Democratic	8994	93	1.03
Kevin McCarthy				
House Minority Leader	Republican	762	0	0.00
Mitch McConnell				
Senate Majority Leader	Republican	5622	3	0.05
Senate Minority Leader	Republican	2483	47	1.89
Nancy Pelosi				
House Majority Leader	Democratic	5617	28	0.50
House Minority Leader	Democratic	2705	30	1.11
Paul Ryan				
House Majority Leader	Republican	3570	6	0.17

Figure 10: Deficit tweeting by congressional leaders (House and Senate; majority vs. minority roles).

Figure 10 summarizes deficit tweeting among House and Senate leaders in both majority and minority roles. Tweeting patteerns vary by party and institutional position, with the sharpest contrasts appearing in the Senate. Senator Chuck Schumer tweeted about deficits far more often as Majority Leader **(4.0%)** than as Minority Leader **(1.0%)**, while Senator Mitch McConnell showed the reverse, posting more in the minority **(1.9%)** than the majority **(0.05%)**. Notably, in contrast to Seantor Schumer, her Democratic counterpart, Representative Nancy Pelosi displayed higher rates in the minority **(1.1%)** than as Speaker **(0.5%)**. By comparison, Representatives Kevin McCarthy and Paul Ryan registered very low shares overall, with McCarthy not posting a single deficit-related tweet.

5 Legislative Context

Deficit Share Within Each Window (Tweet Level)			
Share of tweets that mention the deficit, conditional on being inside the window (2017–2023).			
Window	Tweets (N)	Deficit (N)	Deficit Rate (%)
All Tweets (Baseline)	3,605,411	25,102	0.7%
Any Legislative Window	1,173,979	10,127	0.9%
IRA / CHIPS Window	165,798	2,615	1.6%
CARES Window	214,136	752	0.4%
PPP–HCE Window	223,499	592	0.3%
COVID Window (CARES or PPP–HCE)	280,492	962	0.3%
TCJA Window	147,835	1,819	1.2%
*Each percentage is **deficit tweets ÷ all tweets in that window**. The baseline row is the unconditional rate for all tweets.*			

Figure 11: Deficit share within legislative windows.

Deficit Share Within Each Window by Party (Tweet Level)				
Conditional share of deficit tweets given the tweet is inside the window.				
		Tweets (N)	Deficit (N)	Deficit Rate (%)
All Tweets (Baseline)	Democratic	2,138,163	13,199	0.6%
	Republican	1,467,248	11,903	0.8%
Any Legislative Window	Democratic	687,875	5,569	0.8%
	Republican	486,104	4,558	0.9%
IRA / CHIPS Window	Democratic	91,767	1,194	1.3%
	Republican	74,031	1,421	1.9%
CARES Window	Democratic	131,549	521	0.4%
	Republican	82,587	231	0.3%
PPP–HCE Window	Democratic	135,059	342	0.3%
	Republican	88,440	250	0.3%
COVID Window (CARES or PPP–HCE)	Democratic	171,383	627	0.4%
	Republican	109,109	335	0.3%
TCJA Window	Democratic	85,463	1,680	2.0%
	Republican	62,372	139	0.2%
*Percentages are conditional on being inside each window; counts are tweet totals **within** the window.*				

Figure 12: Deficit share within legislative windows, by party.

As noted, deficit tweeting is rare across Congress, with only **(0.7%)** of all tweets between 2017 and 2023 mentioning the deficit Figure 11 . Legislative periods – defined by one month before and after bill passage – raise the conditional rate modestly to **(0.9%).** Industrial policy bills – the Inflation Reduction Act (IRA) and CHIPS Act, passed under President Biden – generated the highest levels of deficit tweeting **(1.6%)**

among passed legislation. The Tax Cuts and Jobs Act (2017), passed under President Trump, had a deficit tweet share of **(1.2%)** – higher than baseline, but lower than the IRA/CHIPS.

@fig–deficit-share-by-party decomposes these differences by party. At baseline, Republicans tweet about the deficit more often **(0.8%)** than Democrats **(0.6%)** – a gap that persists across most legislative windows. During the IRA and CHIPS periods, Republicans' tweets accelerated, with deficit mentions at nearly **(2.0%)**, compared to **(1.3%)** among Democrats. However, the TCJA – a partisan GOP tax cut– is a noticeable and stark exception. During these three months, the Democrats' deficit share spikes to **(2.0%)**, while the Republicans' share falls to just **(0.2%)**, reflecting a partisan realignment around a highly divisive bill. In contrast, during the COVID relief period in 2020 – a non-partisan voting context – both parties converged at very low levels **(0.3–0.4%)**.

6 Regression Visualization

The following regression models offer a structured approach to examining how institutional, fiscal, and partisan contexts affect the likelihood of congressional deficit tweeting.

Since the specifications rely heavily on interaction terms, it is useful to provide a framework for their interpretation first. The coefficients are expressed as odds ratios: values above one indicate that Republicans are more likely to tweet about deficits relative to Democrats. In contrast, values below one indicate a narrowing of the GOP-Dem gap. Each model includes majority interaction coefficients of the general form GOP $\times$ Conditioning Variable. These coefficients indicate whether the baseline Republican–Democratic difference changes under specific conditions (legislative windows, presidential control, chamber control, or trifectas).

Legislative Windows and Major Bills

	Odds Ratio	95% CI
GOP × Legislative window	0.88	[0.57, 1.36]
GOP × COVID window	0.65	[0.33, 1.28]
GOP × IRA/CHIPS window	1.24	[0.46, 3.32]
GOP × Bill partisanship (z)	0.82	[0.52, 1.29]
GOP × Fiscal magnitude (z)	1.02	[0.97, 1.07]

Republican–Democrat differences in the odds of tweeting about the deficit across legislative windows and major-bill periods, with member and month fixed effects.

- *Odds ratios with 95% CIs; stars: *** p<0.001, ** p<0.01, * p<0.05.*
- *Member and calendar-month fixed effects; SEs two-way clustered by member and month.*
- *Bill partisanship (z): 1 SD increase in mean bill partisanship score.*
- *Fiscal magnitude (z): 1 SD increase in total 10-year deficit impact.*
- *COVID window and IRA/CHIPS window are month-level indicators.*

Figure 13: Regression 1: Legislative windows.

The baseline model, Figure 13, examines partisan differences in legislative periods without conditioning on institutional control. The non-significant results suggest that, when averaged across both Democratic and Republican presidencies, legislative windows themselves do not systematically impact the partisan gap in deficit tweeting.

Presidential Control and Legislative Windows

	Odds Ratio	95% CI
GOP × Legislative window	0.38***	[0.26, 0.56]
GOP × Minority presidency	1.87*	[1.07, 3.27]
GOP × Legislative window × Minority presidency	4.24***	[2.27, 7.92]
GOP × COVID window	1.03	[0.61, 1.74]
GOP × IRA/CHIPS window	0.78	[0.33, 1.85]
GOP × Bill partisanship (z)	0.51***	[0.41, 0.63]
GOP × Fiscal magnitude (z)	0.99	[0.98, 1.01]

Coefficients are GOP–Dem gaps: outside-leg gap under minority presidency; the legislative-window gap under same-party presidency; and how that gap changes under minority presidency.

- *Odds ratios with 95% CIs; stars: *** p<0.001, ** p<0.01, * p<0.05.*
- *Member and calendar-month fixed effects; SEs two-way clustered by member and month.*
- *Bill partisanship (z) and fiscal magnitude (z) are standardized.*

Figure 14: Regression 2: Majority context (presidency).

Figure 14, which controls for presidential party, substantially changes the results from the preceding specification.

Republicans' odds of tweeting about deficits are nearly double **(OR = 1.87)** when not holding the presidency compared to Democrats under the same condition. In contrast, **GOP × Legislative window (OR = 0.38)** highlights that under a Republican presidency, the partisan gap narrows during legislative months; the odds of the GOP tweeting are **62%** lower than those of Democrats. This pattern aligns with Figure 1, which shows that Republican tweeting contracts while Democratic tweeting spikes as Congress passes the Tax Cuts and Jobs Act in late 2017. However, the three-way interaction coefficient **GOP × Legislative window × Minority presidency** reveals a reversal of this trend **(OR = 4.24)**. Under a Democratic presidency, Republicans' odds of tweeting are more than four times higher than those of Democrats in the same condition.

Government Control and Legislative Windows

	Odds Ratio	95% CI
GOP × Legislative window (GOP trifecta)	0.31***	[0.21, 0.46]
GOP × Legislative window × Dem trifecta	7.00***	[4.13, 11.85]
GOP × Legislative window × GOP Senate+Presidency	0.85	[0.47, 1.54]
GOP × COVID window	1.10	[0.62, 1.95]
GOP × IRA/CHIPS window	0.79	[0.33, 1.86]
GOP × Bill partisanship (z)	0.50***	[0.39, 0.64]
GOP × Fiscal magnitude (z)	0.99	[0.98, 1.01]

Baseline: Democrats vs Republicans during legislative windows under a GOP trifecta. Rows report how that GOP–Dem gap changes under alternative government control states. Pure Democrat shifts and a baseline GOP main effect are omitted by design.

- *Odds ratios with 95% CIs; stars: *** p<0.001, ** p<0.01, * p<0.05.*
- *Member and month fixed effects; SEs two-way clustered by member and month.*
- *Bill partisanship (z) and fiscal magnitude (z) are standardized.*

Figure 15: Regression 3: Majority context — chamber and presidency combinations.

The third model further examines the role of institutional power by including combinations of chamber control. The regression baseline is Democrats, outside legislative windows, when their party controls nothing (i.e., a minority in the House and Senate; does not hold the Presidency).

GOP x Legislative window (OR = 0.31) indicates that during legislative months, when holding a trifecta, Republicans' odds of tweeting are **69%** lower than Democrats. The Tax Cuts and Jobs Act (2017) – a legislative window under a GOP trifecta – is an illustration of this circumstance. Once again, once we condition on a Democratic trifecta, this effect reverses. Relative to the baseline condition (a GOP trifecta), Republicans' odds of deficit tweeting are seven times higher than those of Democrats in the same context. The juxtaposition of these two coefficients **(OR = 0.31)** versus **(OR = 7.00)** illustrates that legislative timing by itself isn't determinative of tweeting gaps – but who governs is. Under Democratic control (i.e., 2021-2023), Republicans accelerate deficit rhetoric during legislative months, while under the GOP trifecta (i.e., 2017-2019), they significantly reduce deficit tweeting frequencies.

6.1 Bill Partisanship and Fiscal Magnitude

Across all three specifications, the coefficients for Fiscal magnitude (z) remain consistently insignificant (~1.0). These coefficients, based on a normalized measure of a given bill's 10-year CBO estimate, suggest that the fiscal consequences of policy do not systematically impact partisan tweeting gaps.

Bill partisanship (z), however, shows significant contractions in the GOP-Dem tweeting gap once institutional control is included **(OR ~ 0.50)**. In other words, for each one-SD increase in a bill's partisanship score, the GOP-Dem gap contracts by roughly half. Notably, because these variables are continuous (ie, z-scores), the coefficients represent slopes rather than a baseline-specific effect. For example, in the context of bill partisanship, if Republicans are twice as likely as Democrats to tweet at **z = 0 (OR = 2.0)**, that gap narrows to parity at **z = 1 (OR = 1.0).**